🐘 + 🐠 MULTI-AGENT AUDIT

7 Stateless Subagents Audited Our Spec

Before writing a single line of production code, we ran our CI/CD architecture through the **Elephant-Goldfish Model (EGM)**—catching rate-limit contradictions, security holes, and framework bloat upfront.

🎯 **Who This Is For**

AI Architects, Staff Engineers, and Tech Leads using AI agents to design and review production systems.

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CONFIRMATION BIAS IN LONG CHATS

Why One Agent Rubber-Stamps Its Own Flaws

Asking the same AI session that wrote your design doc to "review it for bugs" fails for two reasons:

🧠 **Shared Context = Shared Blindspots**

The model remembers why it made compromises earlier in the thread—and rationalizes away missing sections or contradictions.

🌀 **Attention Dilution**

In a 50-turn conversation, architectural invariants get buried in the middle of the context window.

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3 Blockers Caught
Before Line 1 of Code

Spawning fresh, stateless reviewer personas exposed real flaws our primary context missed:

🧮 **1. Rate-Limit Math Contradiction**

The spec promised 30 LLM calls in under 90s (20 RPM)—which violated the provider's 15 RPM free-tier ceiling.

🔒 **2. Fork PR Secret Exposure**

External contributor PRs lack GitHub Actions secrets, meaning live cloud evals would crash every outside contribution.

🔀 **3. Doc-vs-Code Drift**

The roadmap cited a heavyweight third-party framework while the actual codebase used a lightweight native SDK.

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COGNITIVE SEPARATION

Pair Persistent Memory With Stateless Critics

Separate the builder who holds institutional memory from the critics who audit the artifact cold:

🐘 **The Elephant (Persistent Resolver)**

Primary context with full project history. Synthesizes findings, rejects over-engineering, and applies final fixes.

🐠 **The Goldfish (Stateless Auditors)**

Ephemeral subagents spawned with **zero chat history**—forced to judge the design doc purely on what is written.

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7 Lenses That Stress-Test Every Architecture Doc

Each Goldfish persona enforces one non-negotiable engineering standard:

PERSONA	CORE MANDATE
Skeptical Architect	Enforces YAGNI; rejects reviewer bloat (e.g., blocked 3-pass voting)
Naive Reader	Flags unstated assumptions & missing file schemas
Hostile Design Critic	Audits scalability, rate-limit math, and security boundaries
Pragmatic Tech Lead	Enforces Zero-Decision Rule & exit verification commands
Codebase Auditor	Verifies spec-to-code parity & catches framework drift

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The Zoom-Out Gate: Rejecting Reviewer Bloat

Critics love suggesting complexity. An intermediate Skeptical Architect pass keeps the core lean:

🚫 **Rejected by Skeptical Architect**

3-pass majority voting, dynamic token-bucket limiters, and VCR recording fixtures that would triple free-tier API burn.

✅ **Shipped Instead**

Pure sequential pacing (~10–12 RPM), `temperature=0.0`, and graceful `pytest.skip` on fork PRs.

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3 Rules for AI-Assisted Architecture Reviews

Catch systemic flaws before your team writes production code:

- ✓ **Review in a fresh context:** Never let the author session grade its own spec.
- ✓ **Assign adversarial roles:** Separate security, comprehension, and codebase parity checks.
- ✓ **Add a Zoom-Out gate:** Explicitly reject over-engineered suggestions before implementation.

Read the Full EGM Case Study

Inspect the full 10-stage pipeline and audit findings at **avilano.dev** (link in first comment).

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